

Windows Firewall Configuration

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Project Title: Windows Firewall Configuration and Rule Management

Objective

The objective of this project was to understand the practical implementation of Windows Defender Firewall by creating, modifying, testing, and verifying custom firewall rules. The project focused on restricting unauthorized network communication, blocking malicious IP addresses, preventing application internet access, restricting ports, and controlling Remote Desktop Protocol (RDP) connectivity.

Introduction

Windows Defender Firewall is a host-based firewall that monitors inbound and outbound network traffic. It allows administrators to define rules that either permit or deny traffic based on IP addresses, ports, applications, protocols, and services.

Task 1: Blocking a Malicious IP Address

Objective

Prevent communication with a suspicious external IP address (65.61.137.117).

Procedure

- Created a custom inbound firewall rule.
- Added the IP address **65.61.137.117** as a blocked remote address.
- Applied the rule to all network profiles (Domain, Private, and Public).

Verification

The connection was tested using the Ping command:

```
ping 65.61.137.117
```

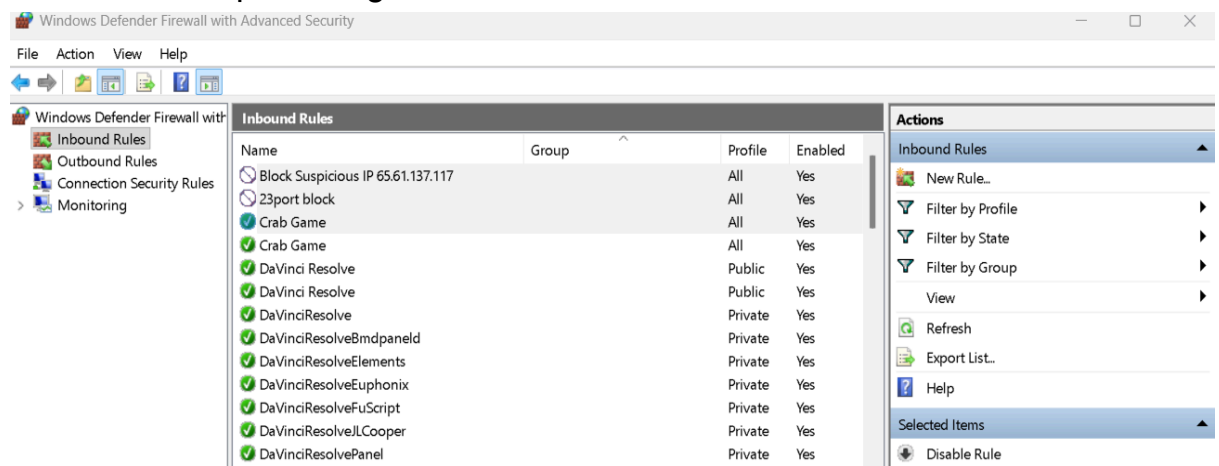
The system returned:

- General failure
- 100% packet loss

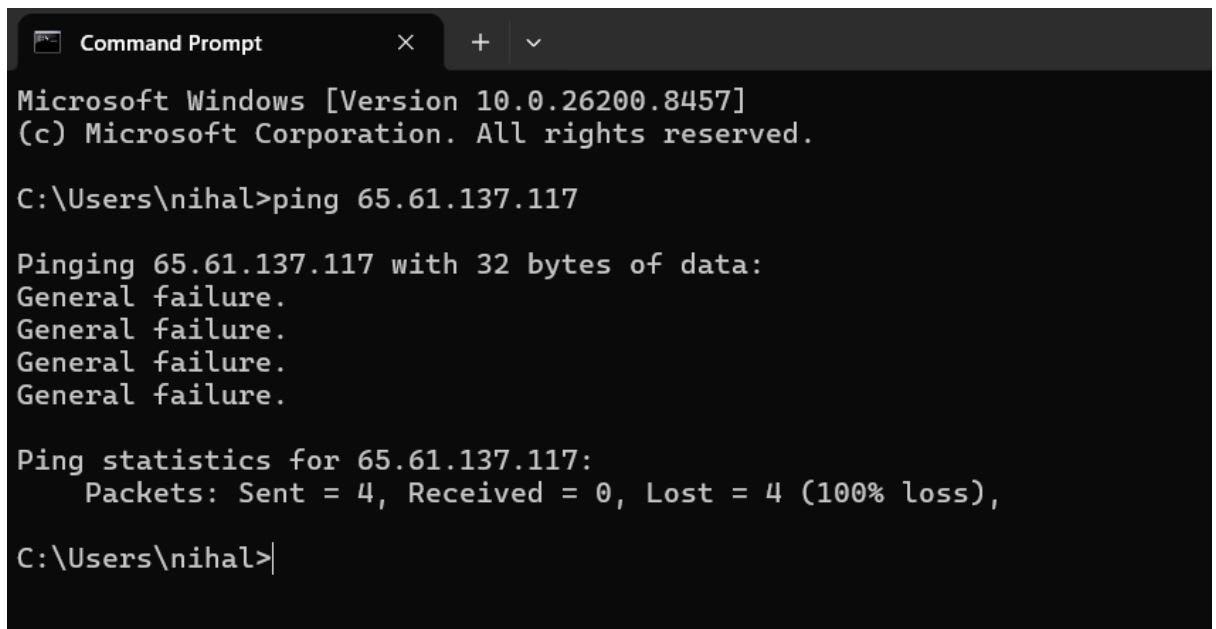
This confirmed that the firewall successfully blocked communication with the specified IP address.

Screenshot

Firewall Rule: ip blocking in



Execution Proof: ip blocking execution

A screenshot of a Windows Command Prompt window. The title bar says "Command Prompt". The text inside shows the Windows version (10.0.26200.8457) and the user's location (C:\Users\nihal). The user has entered the command "ping 65.61.137.117". The output shows four "General failure." messages and a summary: "Ping statistics for 65.61.137.117: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)".

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nihal>ping 65.61.137.117

Pinging 65.61.137.117 with 32 bytes of data:
General failure.
General failure.
General failure.
General failure.

Ping statistics for 65.61.137.117:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\nihal>
```

Task 2: Blocking Google Chrome Internet Access

Objective

Prevent Google Chrome from accessing the internet while leaving other applications unaffected.

Procedure

- Created an outbound firewall rule.
- Selected the Chrome executable.
- Configured the rule to block outbound network traffic.

Verification

After enabling the rule:

Chrome displayed:

Your Internet access is blocked.

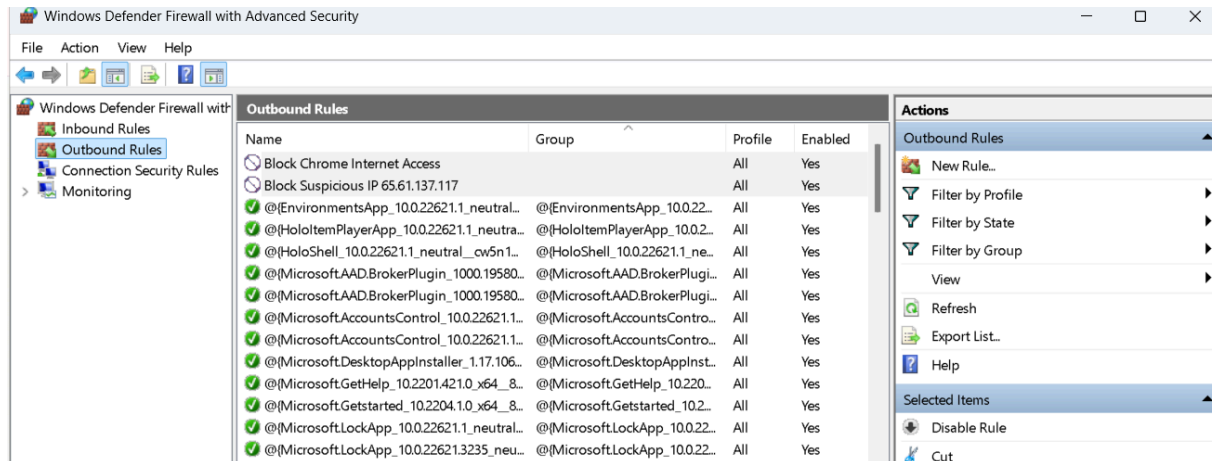
ERR_NETWORK_ACCESS_DENIED

Other applications continued functioning normally.

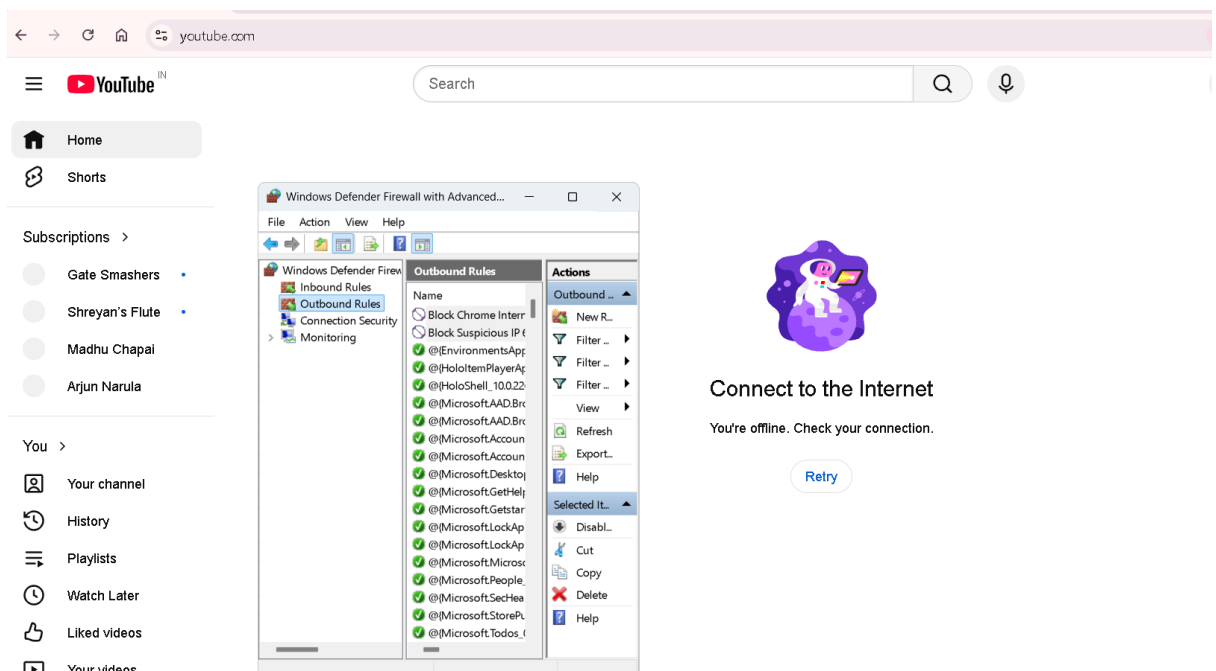
This verified that only Google Chrome was restricted.

Screenshot

- **Firewall Rule:** blocked chrome from accessing internet



- **Execution Proof:** chrome execution



Task 3: Blocking TCP Port 23 (Telnet)

Objective

Disable Telnet communication by blocking TCP Port 23.

Procedure

- Created a custom inbound firewall rule.
- Selected TCP protocol.
- Specified Port 23.
- Configured the rule to block incoming traffic.

Verification

Telnet connectivity was tested using:

telnet demo.testfire.net 23

The result:

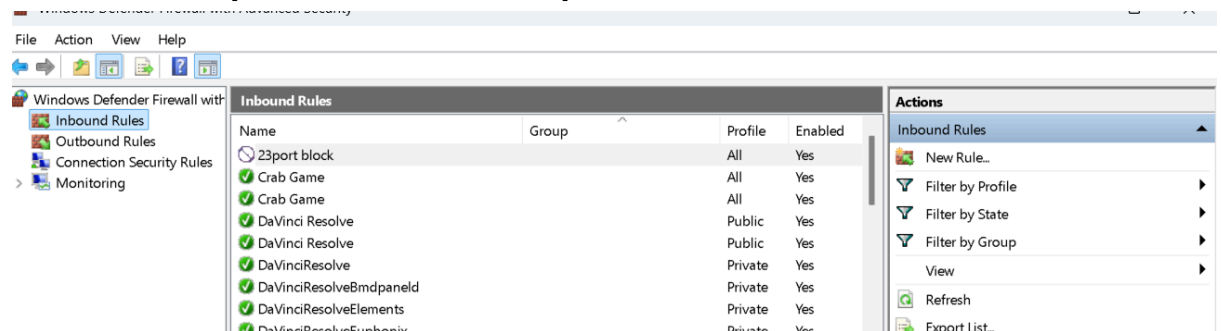
Could not open connection to the host

Connect failed

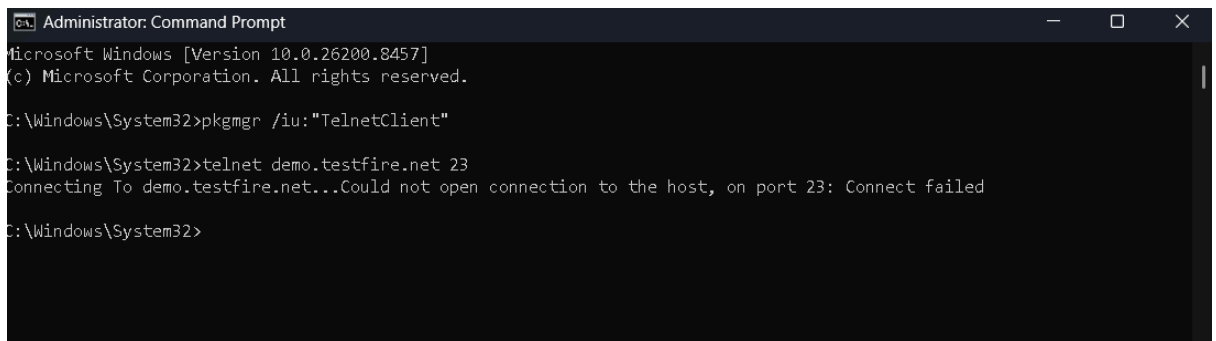
The firewall successfully prevented Telnet communication.

Screenshot

- **Firewall Rule: port 23 rule creation proof**



- **Execution Proof: port23 execution proof**



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>pkgmgr /iu:"TelnetClient"

C:\Windows\System32>telnet demo.testfire.net 23
Connecting To demo.testfire.net...Could not open connection to the host, on port 23: Connect failed

C:\Windows\System32>
```

Task 4: Blocking TCP Port 80 (HTTP)

Objective

Prevent unsecured HTTP communication over Port 80.

Procedure

- Created an outbound firewall rule.
- Selected TCP protocol.
- Blocked Port 80.

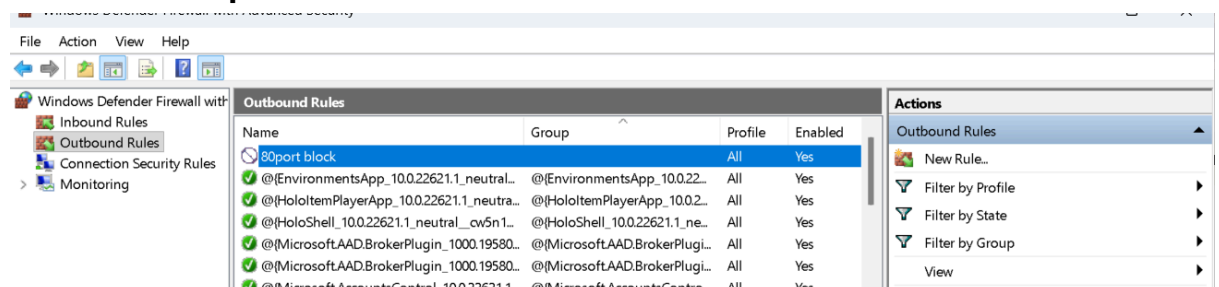
Verification

Port verification was performed.

The firewall rule appeared successfully under Outbound Rules.

Screenshot

- **Firewall Rule: port80 in firewall**



- **Verification: checking port80**



Your Internet access is blocked

Firewall or antivirus software may have blocked the connection.

Try:

- Checking the connection
- [Checking firewall and antivirus configurations](#)
- [Running Windows Network Diagnostics](#)

ERR_NETWORK_ACCESS_DENIED

Details

Task 5: Creating a Restricted Remote Desktop Rule(RDP)

Objective

Allow Remote Desktop (RDP) access only from trusted devices while preventing unauthorized access.

Procedure

- Created a custom inbound firewall rule.
- Allowed RDP traffic only from the specified trusted router/network.
- Applied the rule to all required profiles.

Verification

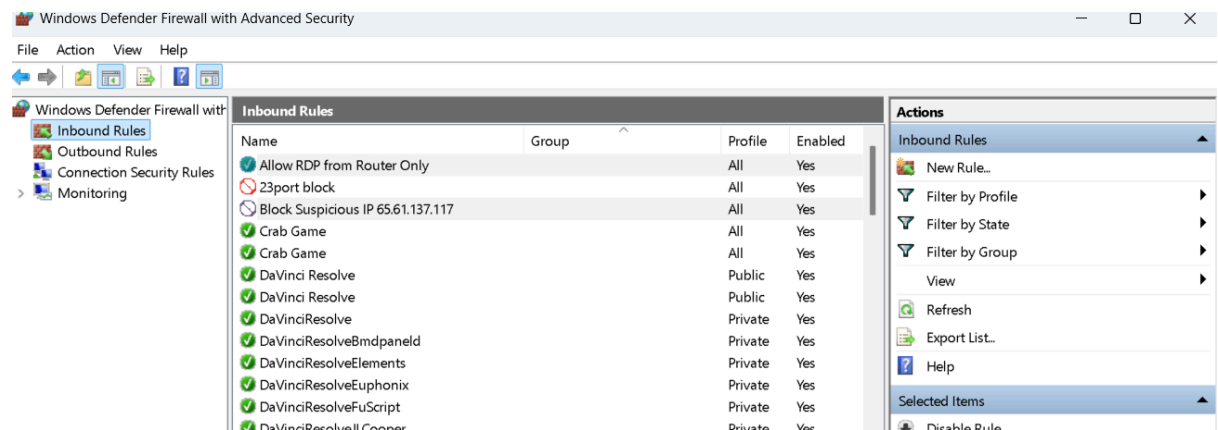
Network connections were checked using:

```
netstat -an | findstr 3389
```

The configured RDP restrictions were successfully applied.

Screenshot

- **Firewall Rule: RDP access to router in firewall**



- **Verification: RDP execution proof**

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\nihal>netstat -an | findstr 3389

C:\Users\nihal>netstat -an | findstr 3389

C:\Users\nihal>
```

Observations

- Firewall rules immediately affected network traffic.
- Application-based rules can selectively restrict internet access.
- IP-based filtering effectively blocks communication with suspicious hosts.
- Port-based filtering prevents insecure protocols such as Telnet.
- RDP access can be limited to trusted networks for improved security.
- Verification using command-line tools is essential to confirm firewall functionality.

Challenges Faced

- Understanding the difference between inbound and outbound rules.
- Identifying the correct ports for different network services.
- Testing firewall rules without disrupting legitimate connectivity.
- Verifying whether rules were correctly enforced using command-line tools.

Conclusion

This project provided practical experience in configuring Windows Defender Firewall to secure a Windows endpoint. Multiple security policies—including IP blocking, application restriction, port filtering, and RDP access control—were successfully implemented and verified. The project demonstrated how host-based firewalls are used in enterprise environments to reduce attack surfaces, prevent unauthorized access, and enforce organizational security policies.